

names are on the hunt for next-generation memory. While this race has been



in the running for well over a decade, it's only recently that a headway has been made in the implementation of these technologies. Giants such as IBM, Infineon and Motorola are in the lead, being specifically involved in the research of a new form of solid-state memory called Magnetoresistive RAM (MRAM). This type of RAM uses principles of alignment of magnetic particles as the basis for storing zeroes and ones in a digital system. With the advancement of magnetic sensing technologies (that are, incidentally, also used in hard disk drives), smaller magnetic fields are capable of being sensed with far greater accuracy than before, allowing for greater densities in storage system. (See box, 'MRAM: Saying Bye to Bootup?')

Another new solid-state memory

technology is Ferroelectric RAM (FRAM). Though it sounds similar to MRAM, this technology uses principles of electric currents to change the polarity of ceramic crystals (and not metallic materials, as the name suggests), which is the principle element for storing digital information. The advantage is that it retains data even after the power is switched off. These crystals also have significantly lower power requirements, making them ideal for mobile consumer devices. Recent



developments in this field have improved the number of times these chips can store data with lower manufacturing costs and increased densities. David Boudurant of Ramtron says that they are only now beginning to evaluate what they can accomplish from the enhanced manufacturing processes.

One very interesting technology that spells volumes for solid-state storage is 3D memory. Until now, the fabrication process of silicon-based memory consisted of layers of microscopic components being fabricated on two-

dimensional surfaces. With 3D memory, scientists have found a way to extend the fabrication process such that the microscopic components (such as transistors and capacitors) can be stacked on top of one another resulting in significantly higher storage densities per unit area of fabricated silicon. This directly spells lower manufacturing costs per megabyte. This concept has been around

for about a decade, but only now when fabrication technologies have been so vastly improved is it viable to put it to practise. This is possible due to a process called chemical mechanical planarisation where silicon surfaces are prepared for vertical stacking of microscopic components. A form of this process is already being used to manufacture flat panel displays. Prototypes of this technology have demonstrated speeds of the order of milliseconds—faster than today's hard disks, but still way slower than



MRAM: Saying Bye to Bootup?

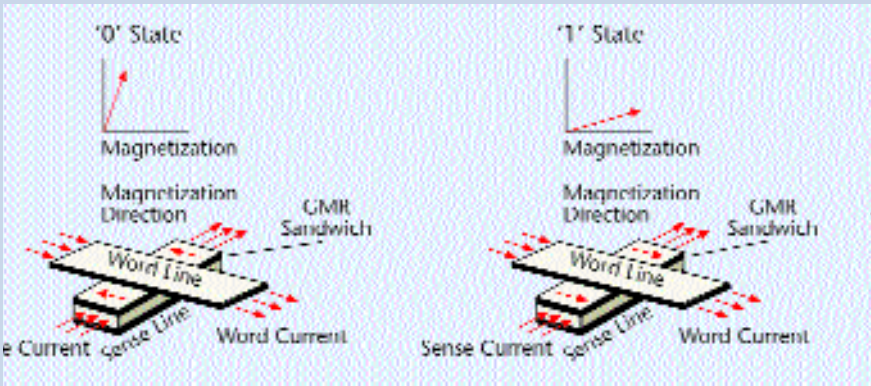
This type of RAM was born out of the same cast as the hard disk drive. Using a phenomenon called Giant Magnetoresistance (GMR), which was first developed by IBM, very small magnetic fields can be stored and sensed reliably and quickly.

MRAM uses the same principle where similar types of GMR materials are used to fabricate arrays of microscopically small magnetic elements. The orientation of the magnetic field in these elements (North or South) determines whether it is carrying a digital zero or one. Each GMR cell is a sandwich made of special metal-

lic materials (like iron and chromium). In an MRAM cell, there are parallel rows of microscopic wires above and below the magnetic sandwiches. The ones above run parallel to the sandwich while the ones below run perpendicular to it, looking much like the weft and warp of a fabric loom. In its normal state, each magnetic element will have an inherent charge in it. If a current is passed through the neighbouring wires, this magnetic orientation can be changed and thus the data can be modified and also written into the element.

While reading data from these elements, a current is passed through the sandwich and the magnetic resistance is measured—a low resistance signifies a zero while high resistance signifies a one. This process of reading does not disturb the magnetic condition of the cell.

The inherent advantage with MRAM is that since it is based on the principle of magnetism, it retains whatever information is contained in it even after power is switched off. Additionally, unlike conventional DRAM, these elements do not need an electric charge to retain the information—it is only needed during read-write processes. Hence, power consumption of these devices is fractional compared to that of DRAM, which needs to be continually refreshed in order to retain data. Finally, with the absence of the delays associated with transferring electricity between the storage elements for retaining the information, MRAM is expected to be up to 30 times faster than DRAM. With these characteristics, MRAM is poised to give rise to a new breed of instant-on computing devices.



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